"Basic & Simplification Questions"

1. The value of i^{2014} is
(a) i (b) $-i$
(c) 1 (d) -1

2. If $a < 0, b > 0$, then $\sqrt{a} \cdot \sqrt{b}$ is equal to
(a) $-\sqrt{|a| \cdot b}$ (b) $\sqrt{|a| \cdot b} \cdot i$
(c) $\sqrt{|a| b}$ (d) None of these

3. The value of the sum $\sum_{n=1}^{13}(i^n + i^{n+1})$, where $i = \sqrt{-1}$ is
(a) i (b) $i - 1$
(c) $-i$ (d) 0

4. The value of $\frac{2^n}{(1+i)^{2n}} + \frac{(1+i)^{2n}}{2^n}, n \in I$, is equal to
(a) 0 (b) 2
(c) $\{1 + (-1)^n\} \cdot i^n$ (d) None of these

5. If $i = \sqrt{-1}$, then the number of values of $i^n + i^{-n}$ for different $n \in I$ is
(a) 3 (b) 2
(c) 4 (d) 1

6. If $z = (-5i)\left(\frac{1}{8}i\right)$, then Im (z) is equal to
(a) 1 (b) 0
(c) -1 (d) None of these

7. If $4x + i(3x - y) = 3 + i(-6)$,
where x and y are real numbers, then t of
x and y respectively are
(a) 3,33 (b) $\frac{3}{5}, \frac{33}{5}$
(c) $\frac{3}{4}, \frac{33}{4}$ (d) N se

8. The value of $3(7 + i7$ is
(a) $15 + 27i$ $28i$
(c) $14 - 28i$ $14 + 23i$

9. The value of $\quad 3 \quad 3) + \left(4 + i\frac{1}{3}\right)] - \left(-\frac{4}{3} + i\right)$ is
(a) $\frac{5}{3} - i\frac{17}{3}$ (b) $\frac{17}{3} - i\frac{5}{3}$
(c) $\frac{17}{3} + i\frac{5}{3}$ (d) $\frac{17}{5} - i\frac{4}{3}$

10. The value of $\frac{z_1}{z_2}$, where $z_1 = 2 + 3i$ and $z_2 = 1 + 2i$, is
(a) $\frac{8}{5} + \frac{1}{5}i$ (b) $\frac{8}{5} - \frac{1}{5}i$
(c) $\frac{1}{5} - \frac{8}{5}i$ (d) None of these

11. If $z + z^3 = 0$, then which of the following must be true on the complex plane?
(a) Re $(z) < 0$ (b) Re $(z) = 0$
(c) lm $(z) = 0$ (d) $z^4 = 1$

12. The value of i^i is
(a) $e^{-\pi/2}$ (b) $e^{\pi/2}$
(c) $e^{\pi/4}$ (d) None of these

13. The value of $\frac{i^{592}+i^{590}+i^{588}+i^{586}+i^{584}}{i^{582}+i^{580}+i^{578}+i^{576}+i^{574}} - 1$ is
(a) -1 (b) -2
(c) -3 (d) -4

14. $i^{57} + \frac{1}{i^{125}}$, when si he value
(a) 0
(c) $-2i$

15. $i^n + i^{n+1}$ i ^{3} is equal to
(a) 1 (b) -1
(c) 0 (d) None of these

16. $1 +$ $+ i^6 + \cdots + i^{2n}$ is
i e (b) negative
(d) Can't be determined

7. The value of $\left[i^{19} + \left(\frac{1}{i}\right)^{25}\right]^2$ is
a) 4 (b) -4
(c) 2 (d) -2

18. If n is any positive integer, then the value of $\frac{i^{4n+1} - i^{4n-1}}{2}$ equals
(a) 1 (b) -1
(c) i (d) $-i$

19. For a positive integer n, the expression $(1 - i)^n \left(1 - \frac{1}{i}\right)^n$ equals
(a) 0 (b) $2i^n$
(c) 2^n (d) 4^n

20. If the number $\frac{(1-i)^n}{(1+i)^{n-2}}$ is real and positive, then n is
(a) any integer (b) 2λ
(c) $4\lambda + 1$ (d) None of these

21. The smallest positive integer n for which $\left(\frac{1+i}{1-i}\right)^n = -1$ is
(a) 1 (b) 2
(c) 3 (d) 4

22. The smallest positive number n for which $(1+i)^{2n} = (1-i)^{2n}$ is

(a) 4 (b) 8
(c) 2 (d) 12

23. The complex number $\frac{1+2i}{1-i}$ lies in the

(a) I quadrant (b) II quadrant
(c) III quadrant (d) IV quadrant

24. The value of $\frac{(1-i)^3}{1-i^3}$ is equal to

(a) i (b) -1
(c) 1 (d) -2

25. $\left(\frac{1+i}{1-i}\right)^2 + \left(\frac{1-i}{1+i}\right)^2$ is equal to

(a) $2i$ (b) $-2i$
(c) -2 (d) 2

26. The value of Re $\left[\frac{(1+i)^2}{3-i}\right]$ is equal to

(a) $-\frac{1}{5}$ (b) $\frac{1}{5}$
(c) $\frac{1}{10}$ (d) $-\frac{1}{10}$

27. If $\begin{vmatrix} 6i & -3i & 1 \\ 4 & 3i & -1 \\ 20 & 3 & i \end{vmatrix} = x + iy$, then

(a) $x = 3, y = 1$ (b) $x = 1, y = 3$
(c) $x = 0, y = 3$ (d) $x = 0, y = 0$

28. If z is a complex number such that |
Re $(z) = 0$, then

(a) Re $(z^2) = 0$ (b) Im (z^2)
(c) Re $(z^2) =$ Im (z^2) (d) None o

29. If $(x+iy)^{1/3} = a + ib$, t $\frac{\ }{b}$ is equal to

(a) $2(a^2 - b^2)$ (b))
(c) $8(a^2 - b^2)$ (of these

30. If $\left(\frac{1+i}{1-i}\right)^3 -$ $+ iy$, then (x, y) is equal to

(a) $(0,2)$ (b) $(-2,0)$
(c) $(0,-2)$ (d) None of these

31. The value of $\left(\frac{1+i}{\sqrt{2}}\right)^8 + \left(\frac{1-i}{\sqrt{2}}\right)^8$ is equal to

(a) 4 (b) 6
(c) 8 (d) 2

32. If $z = x - iy$ and $z^{1/3} = p + iq$, then $\left(\frac{x}{p} + \frac{y}{q}\right)/(p^2 + q^2)$ is equal to

(a) 1 (b) -1
(c) 2 (d) -2

33. If $\left(\frac{1+i}{1-i}\right)^x = 1$, then

(a) $x = 4n$, where n is any positive integer
(b) $x = 2n$, where n is any positive integer
(c) $x = 4n + 1$, where n is any positive integer
(d) $x = 2n + 1$, where n is any positive integer

34. If $z_1 = 2\sqrt{2}(1+i)$ and $z_2 = 1 + i\sqrt{3}$, then $z_1^2 z_2^3$ is equal to

(a) $128i$ (b) $64i$
(c) $-64i$ (d) *None of these*

35. The value of $(1+i)^3 + (1-i)^3$ is equal to

(a) 1 (b) -2
(c) 0

36. The sequence S $+ 3i^3 + 4i^4 + \cdots$ upto 100 terms simplif here $i = \sqrt{-1}$

(a) $50(1-i)$ (b) $25i$
(c) $25(1+i)$ (d) $100(1-i)$

37. A v θ for which $\frac{2+3i\sin\theta}{1-2i\sin\theta}$ is purely

(b) $\frac{\pi}{6}$
in$^{-1}\left(\frac{\sqrt{3}}{4}\right)$ (d) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

8. The real values of x and y for which the following equality hold, are respectively

$$(x^4 + 2xi) - (3x^2 + iy) = (3 - 5i) + (1 + 2iy)$$

(a) $2,3$ or $-2,1/3$ (b) $1,3$ or $-1,1/3$
(c) $2,1/3$ or $-2,3$ (d) None of these

39. Let $\left(-2 - \frac{1}{3}i\right)^3 = \frac{x+iy}{27}$ $(i = \sqrt{-1})$, where x and y are real numbers, then $y - x$ equals

(a) 91 (b) 85
(c) -85 (d) -91

40. If $a = \cos\theta + i\sin\theta$, then $\frac{1+a}{1-a}$ is equal to

(a) $i\cot\frac{\theta}{2}$ (b) $i\tan\frac{\theta}{2}$
(c) $i\cos\frac{\theta}{2}$ (d) $i\operatorname{cosec}\frac{\theta}{2}$

41. Real part of $\frac{1}{1-\cos\theta + i\sin\theta}$ is

(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
(c) $\frac{1}{2}\tan\theta/2$ (d) 2

42. The value of $x^4 + 9x^3 + 35x^2 - x + 4$ for $x = -5 + 2\sqrt{-4}$ is

(a) 0 (b) -160
(c) 160 (d) -164

"Conjugate of Complex Number"

43. If $z_1 = 9y^2 - 4 - 10$ ix and $z_2 = 8y^2 + 20i$, where $z_1 = \bar{z}_2$, then $z = x + iy$ is equal to
(a) $-2 + 2i$ (b) $-2 \pm 2i$
(c) $-2 \pm i$ (d) None of the above

44. If $(1 + i)z = (1 - i)\bar{z}$, then z is
(a) $x(1 - i), x \in R$ (b) $x(1 + i), x \in R$
(c) $\frac{x}{1+i}, x \in R^+$ (d) None of the above

45. If $z = \left(\frac{\sqrt{3}}{2} + \frac{i}{2}\right)^5 + \left(\frac{\sqrt{3}}{2} - \frac{i}{2}\right)^5$, then
(a) Re $(z) = 0$
(b) Im $(z) = 0$
(c) Re $(z) > 0$, Im $(z) > 0$
(d) Re $(z) > 0$, Im $(z) < 0$

46. Statement **I** : $3 + ix^2y$ and $x^2 + y + 4i$ are complex conjugate numbers, then $x^2 + y^2 = 4$.

Statement II : If sum and product of two complex numbers is real, then they are conjugate co number.

Which of the above statement(s) is/are

(a) Only I (b) Only II
(c) Both I and II (d) Neithe

47. If $a^2 + b^2 = 1$, then $\frac{(1+b}{(1}$ ual to
(a) 1 (
(c) $b + ia$ b

48. If $z = x + i$ I quadrant, then $\frac{\bar{z}}{z}$ also lies in III quad
(a) $x > y > 0$ (b) $x < y < 0$
(c) $y < x < 0$ (d) $y > x > 0$

49. The conjugate complex number of $\frac{2-i}{(1-2i)^2}$ is
(a) $\frac{2}{25} + \frac{11}{25}i$ (b) $\frac{2}{25} - \frac{11}{25}i$
(c) $-\frac{2}{25} + \frac{11}{25}i$ (d) $-\frac{2}{25} - \frac{11}{25}i$

50. The conjugate of a complex number is $\frac{1}{i-1}$. Then, that complex number is
(a) $-\frac{1}{i-1}$ (b) $\frac{1}{i+1}$
(c) $-\frac{1}{i+1}$ (d) $\frac{1}{i-1}$

51. If the conjugate of $(x + iy)(1 - 2i)$ is $1 + i$, then
(a) $x - iy = \frac{1-i}{1-2i}$ (b) $x + iy = \frac{1-i}{1-2i}$
(c) $x = \frac{1}{5}$ (d) $x = -\frac{1}{5}$

52. If $z = \frac{4}{1-i}$, then $\bar{z}$ is (where, $\bar{z}$ is complex conjugate of z)
(a) $2(1 + i)$ (b) (1
(c) $\frac{2}{1-i}$ (d) 4

53. The complex num $+ i\cos\ 2x$ and $\cos\ x - i\sin\ 2x$ ar e to each other for
(a) $x = n\pi$ (b) $x = \left(n + \frac{1}{2}\right)\pi$
(c) $x =$ (d) No value of x

" s of Complex Number"

a complex number satisfying the n $|z + 1| = z + 2(1 + i)$, then z is
$\frac{}{2}(1 + 4i)$ (b) $\frac{1}{2}(3 + 4i)$
c) $\frac{1}{2}(1 - 4i)$ (d) $\frac{1}{2}(3 - 4i)$

55. If $z = 1 + i\tan\ \alpha$, where $\pi < \alpha < \frac{3\pi}{2}$, then $|z|$ is equal to
(a) $\sec\ \alpha$ (b) $-\sec\ \alpha$
(c) $\text{cosec}\ \alpha$ (d) None of the above

56. The complex number z satisfies $z + |z| = 2 + 8i$.
Then, the value of $|z|$ is
(a) 10 (b) 13
(c) 17 (d) 23

57. Consider two complex numbers α and β as

$\alpha = \left(\frac{a+bi}{a-bi}\right)^2 + \left(\frac{a-bi}{a+bi}\right)^2$, where $a, b \in R$ and

$\beta = \frac{z-1}{z+1}$, where $|z| = 1$, then
(a) both α and β are purely real
(b) both α and β are purely imaginary
(c) α is purely real and β is purely imaginary
(d) β is purely real and α is purely imaginary

58. If $x - iy = \sqrt{\frac{a-ib}{c-id}}$, then $(x^2+y^2)^2$ is equal to

(a) $\frac{a^2-b^2}{c^2-d^2}$ (b) $\frac{a^2+b^2}{c^2+d^2}$
(c) $\frac{a^2+b^2}{c^2-d^2}$ (d) None of these

59. If α, β are two complex numbers, then $|\alpha|^2 + |\beta|^2$ is equal to

(a) $\frac{1}{2}(|\alpha+\beta|^2 - |\alpha-\beta|^2)$
(b) $\frac{1}{2}(|\alpha+\beta|^2 + |\alpha-\beta|^2)$
(c) $|\alpha+\beta|^2 + |\alpha-\beta|^2$
(d) None of these

60. Let z be a complex number satisfying the equation $(z^3+3)^2 = -16$, then $|z|$ has the value equal to

(a) $5^{1/2}$ (b) $5^{1/3}$
(c) $5^{2/3}$ (d) 5

61. If z is a complex number satisfying $z^4+z^3+2z^2+z+1=0$, then $|z|$ is equal to

(a) $\frac{1}{2}$ (b) $\frac{3}{4}$
(c) 1 (d) None of these

62. If $\frac{1-i\alpha}{1+i\alpha} = A + iB$, then A^2+B^2 equals

(a) 1 (b) α^2
(c) -1 (d) $-\alpha^2$

63. If z_1 and z_2 are two complex numbe satisfying the equation $\left|\frac{z_1+iz_2}{z_1-iz_2}\right| = 1$, -

(a) purely real (b) of unit
(c) purely imaginary (d) N above

64. If $8iz^3 + 12z^2 - 18z$ then

(a) $|z| = \frac{3}{2}$ $\frac{2}{3}$
(c) $|z| = 1$ $|z| = \frac{3}{4}$

65. The value of $\left|\frac{1+i\sqrt{3}}{\left(1+\frac{1}{i+1}\right)^2}\right|$ is

(a) 20 (b) 9
(c) $\frac{5}{4}$ (d) $\frac{4}{5}$

66. If $z_1 = \sqrt{2}\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$ and $z_2 = \sqrt{3}\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$, then $|z_1 z_2|$ is

(a) 6 (b) $\sqrt{6}$
(c) $\sqrt{2}$ (d) $\sqrt{3}$

67. $\left|(1+i)\left(\frac{2+i}{3+i}\right)\right|$ is equal to

(a) $-\frac{1}{2}$ (b) $\frac{1}{2}$
(c) 1 (d) -1

68. If $\frac{1-ix}{1+ix} = a - ib$ and $a^2+b^2=1$, where a and b are real, then x is equal to

(a) $\frac{2a}{(1+a)^2+b^2}$ (b) $\frac{2b}{(1+a)^2+b^2}$
(c) $\frac{2a}{(1+b)^2+a^2}$ (d) $\frac{2b}{(1+b)^2+a^2}$

69. The modulus of $\frac{(3+2i)^2}{(4-3i)}$ is

(a) $\frac{13}{5}$ (b) 11
(c) $\frac{9}{5}$

70. If $x + iy$ $\frac{u+}{}$ x^2+y^2 is equal to

(a) 1 (b) -1
(c) 0 (d) None of these

71. If z mplex number, then

$^2|$ $|z|$ (b) $|z^2| = |z|^2$
$|$ $|z|^2$ (d) $|z^2| \geq |z|^2$

f $z_1 \neq z_2$ and $|z_1+z_2| = \left|\frac{1}{z_1}+\frac{1}{z_2}\right|$, then

a) atleast one of z_1, z_2 is unimodular
(b) $z_1 \cdot z_2$ is unimodular
(c) $z_1 \cdot z_2$ is non-unimodular
(d) None of the above

73. If z satisfies the equation $|z| - z = 1 + 2i$, then z is equal to

(a) $\frac{3}{2} + 2i$ (b) $\frac{3}{2} - 2i$
(c) $2 - \left(\frac{3}{2}\right)i$ (d) $2 + \left(\frac{3}{2}\right)i$

74. If $z = \frac{(\sqrt{3}+i)^3(3i+4)^2}{(8+6i)^2}$, then $|z|$ is equal to

(a) 8 (b) 2
(c) 5 (d) None of these

75. If z_1, z_2 and z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = \left|\frac{1}{z_1}+\frac{1}{z_2}+\frac{1}{z_3}\right| = 1$, then $|z_1+z_2+z_3|$ is

(a) 3 (b) 1
(c) greater than 3 (d) less than 1

76. If $(\sqrt{3}i+1)^{100} = 2^{99}(a+ib)$, then a^2+b^2 is equal to

(a) 4 (b) 3
(c) 2 (d) 0

77. If $w = \alpha + i\beta$, where $\beta \neq 0$ and $z \neq 1$, satisfies the condition that $\left(\frac{w - \bar{w}z}{1-z}\right)$ is purely real, then the set of values of z is

(a) $|z| = 1, z \neq 2$ (b) $|z| = 1$ and $z \neq 1$

(c) $z = \bar{z}$ (d) None of these

78. If $\frac{z-\alpha}{z+\alpha}(\alpha \in R)$ is a purely imaginary number and $|z| = 2$, then a value of α is

(a) $\sqrt{2}$ (b) $\frac{1}{2}$

(c) 1 (d) 2

79. If Re $(z^2) = 0, |z| = 2$, then

(a) $z = \mp\sqrt{2} \pm i\sqrt{2}$ (b) $z = \mp\sqrt{3} \pm i\sqrt{3}$

(c) $z = \mp\sqrt{3} \pm i\sqrt{2}$ (d) None of these

80. If $\frac{z-1}{z+1}$ is a purely imaginary number $(z \neq -1)$, then the value of $|z|$ is

(a) -1 (b) 1

(c) 2 (d) -2

81. If $z + \sqrt{2}|z + 1| + i = 0$, then z equals

(a) $2 + i$ (b) $-2 + i$

(c) $-\frac{1}{2} + i$ (d) $-2 - i$

82. If $f(z) = \frac{7-z}{1-z^2}$, where $z = 1 + 2i$, then $|f(z)|$ is

(a) $\frac{|z|}{2}$ (b) $|z|$

(c) $2|z|$ (d) None of these

"Multiplicative Inverse

83. The multiplicative inverse of

(a) $\frac{4}{25} - \frac{3i}{25}$ (b) $\frac{4}{25}$

(c) $\frac{4}{16} + \frac{3i}{25}$ (d) ese

84. If $z = 1 + i$, ltiplicative inverse of z^2 is

(a) $1 - i$ (b) $\frac{i}{2}$

(c) $-\frac{i}{2}$ (d) $2i$

85. The multiplicative inverse of $(6 + 5i)^2$ is

(a) $\frac{11}{60} - \frac{60}{61}i$ (b) $\frac{11}{61} - \frac{60}{61}i$

(c) $\frac{9}{61} - \frac{60}{61}i$ (d) None of these

86. The multiplicative inverse of $\frac{3+4i}{4-5i}$ is

(a) $-\frac{8}{25} + \frac{31}{25}i$ (b) $\frac{8}{25} - \frac{31}{25}i$

(c) $-\frac{8}{25} - \frac{31}{25}i$ (d) None of these

"Square Root of Complex Number"

87. The value of $\sqrt{-8 - 6i}$ is equal to

(a) $1 \pm 3i$ (b) $\pm(1 - 3i)$

(c) $\pm(1 + 3i)$ (d) $\pm(3 - i)$

88. The value of $(4 + 3\sqrt{-20})^{1/2} + (4 - 3\sqrt{-20})^{1/2}$ is

(a) ± 6 (b) 0

(c) $\pm\sqrt{5}$ (d) ± 3

89. Consider the following s nts

If $x, y \in R$, then

I. $\sqrt{x + iy} + \sqrt{x -}$ $\left(\sqrt{x^2 + y^2} + x\right)$

II. $\sqrt{x}$ $= i\sqrt{2\left(\sqrt{x^2 + y^2} - x\right)}$

Which above statement(s) is/are correct ?

l (b) Only II

th I and II (d) Neither I nor II

$\sqrt{2i}$ equals

a) $1 + i$ (b) $1 - i$

(c) $-\sqrt{2i}$ (d) None of these

91. The value of $|\sqrt{2i} - \sqrt{-2i}|$ is

(a) 2 (b) $\sqrt{2}$

(c) 0 (d) $2\sqrt{2}$

92. The square roots of $-2 + 2\sqrt{3}i$ are

(a) $\pm(1 + \sqrt{3}i)$ (b) $\pm(1 - \sqrt{3}i)$

(c) $\pm(-1 + \sqrt{3}i)$ (d) None of these

93. The imaginary part of $(3 + 2\sqrt{-54})^{1/2} - (3 - 2\sqrt{-54})^{1/2}$ can be

(a) $-\sqrt{6}$ (b) $-2\sqrt{6}$

(c) 6 (d) $\sqrt{6}$

94. The square root of $7 + 24i$ are

(a) $\pm(3 + 4i)$ (b) $\pm(3 - 4i)$

(c) $\pm(4 + 3i)$ (d) $\pm(4 - 3i)$

"Argument of Complex Number"

95. The modulus and argument of the complex number $\frac{1+2i}{1-3i}$ is

(a) $\frac{1}{\sqrt{2}}, \frac{3\pi}{4}$
(b) $\frac{1}{\sqrt{2}}, -\frac{3\pi}{4}$
(c) $\frac{1}{2}, \frac{3\pi}{4}$
(d) None of the above

96. Let z and w be two non-zero complex numbers, such that $|z| = |w|$ and $\arg(z) + \arg(w) = \pi$. Then, z is equal to
(a) w
(b) $-w$
(c) $-\bar{w}$
(d) $\bar{w}$

97. If $z = \frac{\sqrt{3}+i}{\sqrt{3}-i}$, then the fundamental amplitude of z is
(a) $-\frac{\pi}{3}$
(b) $\frac{\pi}{3}$
(c) $\frac{\pi}{6}$
(d) None of these

98. If z_1, z_2 and z_3, z_4 are two pairs of conjugate complex numbers, then $\arg\left(\frac{z_1}{z_4}\right) + \arg\left(\frac{z_2}{z_3}\right)$ equals to
(a) 0
(b) $\frac{\pi}{2}$
(c) $\frac{3\pi}{2}$
(d) π

99. If $\frac{1+2i}{2+i} = r(\cos\theta + i\sin\theta)$, then
(a) $r = 1, \theta = \tan^{-1}\frac{3}{4}$
(b) $r = \sqrt{5}, \theta = \tan^{-1}\frac{4}{3}$
(c) $r = 1, \theta = \tan^{-1}\frac{4}{3}$
(d) None of these

100. If z_1 and z_2 are two non-zero c numbers, such that $|z_1 + z_2| = |z$ then $\arg(z_1) - \arg(z_2)$ is equal t
(a) $-\pi$
(b) $-$
(c) 0
(

101. If $z = 1 -$ s α, where $\alpha \in \left(0, \frac{\pi}{2}\right)$, then the modul the principal value of the argument of z are respectively
(a) $\sqrt{2(1-\sin\alpha)}, \left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$
(b) $\sqrt{2(1-\sin\alpha)}, \left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$
(c) $\sqrt{2(1+\sin\alpha)}, \left(\frac{\pi}{4} + \frac{\alpha}{2}\right)$
(d) $\sqrt{2(1+\sin\alpha)}, \left(\frac{\pi}{4} - \frac{\alpha}{2}\right)$

102. The principal argument of the complex number $\frac{(1+i)^5(1+\sqrt{3}i)^2}{2i(-\sqrt{3}+i)}$ is
(a) $\frac{19\pi}{12}$
(b) $-\frac{7\pi}{12}$
(c) $-\frac{5\pi}{12}$
(d) $\frac{5\pi}{12}$

103. The complex number $i + \sqrt{3}$ in polar form can be written as
(a) $\frac{1}{\sqrt{2}}\left(\sin\frac{\pi}{6} + i\cos\frac{\pi}{6}\right)$
(b) $2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$
(c) $\frac{1}{2}\left(\sin\frac{\pi}{6} + i\cos\frac{\pi}{6}\right)$
(d) $4\left(\cos\frac{\pi}{6} + i\cos\frac{\pi}{6}\right)$

104. If $\sqrt{3} + i = (a+ib)(c+id)$, then $\tan^{-1}\frac{b}{a} + \tan^{-1}\frac{d}{c}$ is equal to
(a) $\frac{\pi}{2}$
(b) -
(c) $\frac{\pi}{6}$

105. If z is a com r of unit modulus and argument θ, $\left(\frac{z}{1+\bar{z}}\right)$ equals
(a) $-\theta$
(b) $\frac{\pi}{2} - \theta$
(c) θ
(d) $\pi - \theta$

$(z-1) = \arg(z+3i)$, then $x - 1 : y$ is to
(b) $1:3$
$:2$
(d) $2:3$

"De-Moivre's Theorem"

107. If $x_r = \cos\left(\frac{\pi}{2^r}\right) + i\sin\left(\frac{\pi}{2^r}\right)$, then the value of $x_1 x_2 x_3 \ldots \infty$ is
(a) -1
(b) 1
(c) 0
(d) None of these

108. $\frac{(\cos\theta + i\sin\theta)^4}{(\sin\theta + i\cos\theta)^5}$ is equal to
(a) $\cos\theta - i\sin\theta$
(b) $\cos 9\theta - i\sin 9\theta$
(c) $\sin\theta - i\cos\theta$
(d) $\sin 9\theta - i\cos 9\theta$

109. The expression $\left[\frac{1+\sin\frac{\pi}{8}+i\cos\frac{\pi}{8}}{1+\sin\frac{\pi}{8}-i\cos\frac{\pi}{8}}\right]^8$ is equal to
(a) 1
(b) -1
(c) I
(d) $-i$

110. If $z = \frac{\cos\theta + i\sin\theta}{\cos\theta - i\sin\theta}, \frac{\pi}{4} < \theta < \frac{\pi}{2}$. Then, $\arg(z)$ is
(a) 2θ
(b) $2\theta - \pi$
(c) $\pi + 2\theta$
(d) None of these

111. If $a = \cos\alpha + i\sin\alpha$, $b = \cos\beta + i\sin\beta$, $c = \cos\gamma + i\sin\gamma$ and $\frac{a}{b} + \frac{b}{c} + \frac{c}{a} = 1$, then $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha)$ is equal to

(a) $\frac{3}{2}$ (b) $-\frac{3}{2}$
(c) 0 (d) 1

112. If $2\cos\theta = x + \frac{1}{x}$ and $2\cos\phi = y + \frac{1}{y}$, then

(a) $\frac{x}{y} + \frac{y}{x} = 2\cos(\theta + \phi)$
(b) $x^m y^n + \frac{1}{x^m y^n} = 2\cos(m\theta + n\phi)$
(c) $\frac{x^m}{y^n} + \frac{y^n}{x^m} = 2\cos(m\theta + n\phi)$
(d) $xy + \frac{1}{xy} = 2\cos(\theta - \phi)$

113. The value of $\frac{\left(\sin\frac{\pi}{8} + i\cos\frac{\pi}{8}\right)^8}{\left(\sin\frac{\pi}{8} - i\cos\frac{\pi}{8}\right)^8}$ is

(a) -1 (b) 0
(c) 1 (d) $2i$

114. If $a = \cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}$, then the value of $\left(\frac{1+a}{2}\right)^{3n}$ is

(a) $(-1)^n$ (b) $\frac{(-1)^n}{2^{3n}}$
(c) $\frac{1}{2^{3n}}$ (d) $(-1)^n + 1$

115. The value of $\frac{4(\cos 75° + i\sin 75°)}{0.4(\cos 30° + i\sin 30°)}$ is

(a) $\frac{10}{\sqrt{2}}(1+i)$ (b) $\frac{10}{\sqrt{2}}(1-i)$
(c) $\frac{5}{\sqrt{2}}(1+i)$ (d) None o

116. The value of $(\cos 2\theta$ $^7(\cos 3\theta + i\sin 3\theta)^{-5}$ is $(\cos 4\theta +$ $^{12}(\cos 5\theta + i\sin 5\theta)^{-6}$

(a) $\cos 33\theta + i$
(b) $\cos 33\theta - i$
(c) $\cos 47\theta + i\sin 47\theta$
(d) $\cos 47\theta - i\sin 47\theta$

117. $(\cos 2\theta + i\sin 2\theta)^{-5}(\cos 3\theta - i\sin 3\theta)^6 (\sin\theta - i\cos\theta)^3$ is equal to

(a) $\cos 25\theta + i\sin 25\theta$
(b) $\cos 25\theta - i\sin 25\theta$
(c) $\sin 25\theta + i\cos 25\theta$
(d) $\sin 25\theta - i\cos 25\theta$

118. If $a = \cos 2\alpha + i\sin 2\alpha$, $b = \cos 2\beta + i\sin 2\beta$, $c = \cos 2\gamma + i\sin 2\gamma$ and $d = \cos 2\delta + i\sin 2\delta$, then $\sqrt{abcd} + \frac{1}{\sqrt{abcd}}$ is equal to

(a) $\sqrt{2}\cos(\alpha + \beta + \gamma + \delta)$
(b) $2\cos(\alpha + \beta + \gamma + \delta)$
(c) $\cos(\alpha + \beta + \gamma + \delta)$
(d) None of these

119. The value of $\left(\frac{1 + \sin\frac{2\pi}{9} + i\cos\frac{2\pi}{9}}{1 + \sin\frac{2\pi}{9} - i\cos\frac{2\pi}{9}}\right)^3$ is

(a) $-\frac{1}{2}(\sqrt{3} - i)$ (b) $-\frac{1}{2}(1 - i\sqrt{3})$
(c) $\frac{1}{2}(\sqrt{3} - i)$ (d) $\frac{1}{2}(1 - i\sqrt{3})$

“Cube Ro y”

120. The value of a amp $(i\omega^2)$, where $i = \sqrt{-1}$ and $\omega =$ on-real, is

(a) 0 (b) $\frac{\pi}{2}$
(c) π (d) None of these

121. If $+ 1 = 0$, then the value of x $+ \frac{1}{x^n}\Big)^2$ is

(b) 10
2 (d) None of the above

122. If $3^{49}(x + iy) = \left(\frac{3}{2} + \frac{\sqrt{3}}{2}i\right)^{100}$ and $x = ky$, then k is

(a) $-\frac{1}{3}$ (b) $\sqrt{3}$
(c) $-\sqrt{3}$ (d) $-\frac{1}{\sqrt{3}}$

123. If $z^2 - z + 1 = 0$, then $z^n - z^{-n}$, where n is a multiple of 3 , is

(a) $2(-1)^n$ (b) 0
(c) $(-1)^{n+1}$ (d) None of the above

124. If α is non-real and $\alpha = \sqrt[3]{1}$, then the value of $2^{|1+\alpha+\alpha^2+\alpha^{-2}+\alpha^{-1}|}$ is equal to

(a) 4 (b) 1/2
(c) 2 (d) None of these

125. If ω is a non-real cube root of unity, then the expression $(1-\omega)(1-\omega^2)(1+\omega^4)(1+\omega^8)$ is equal to

(a) 0 (b) 3
(c) 1 (d) 2

126. If ω is a non-real cube root of unity, then $\frac{1+2\omega+3\omega^2}{2+3\omega+\omega^2}+\frac{2+3\omega+\omega^2}{3+\omega+2\omega^2}$ is equal to
(a) -1 (b) 2ω
(c) 0 (d) -2ω

127. If $(\sqrt{3}+i)^n=(\sqrt{3}-i)^n, n\in N$, then least value of n is
(a) 3 (b) 4
(c) 6 (d) None of these

128. If $x^3-1=0$ has the non-real complex roots α,β, then the value of $(1+2\alpha+\beta)^3-(3+3\alpha+5\beta)^3$ is
(a) -7 (b) 6
(c) -5 (d) 0

129. If $z^2-z+1=0$, then the value of

$$\left(z+\frac{1}{z}\right)^2+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)^2$$

$+\cdots+\left(z^{24}+\frac{1}{z^{24}}\right)$ is equal to
(a) 24 (b) 32
(c) 48 (d) None of these

130. When the polynomial $5x^3+Mx+N$ is divided by x^2+x+1, the remainder is value of $(M+N)$ is equal to
(a) -3 (b) 5
(c) -5 (d) 15

131. If $z=-2+2\sqrt{3}i$, then $+2^{4n}$ may be equal to
(a) 2^{2n}
(b) 0
(c) $3\cdot 4^{2n}, n$ is
(d) None of the

132. Statement I: If $x+\frac{1}{x}=1$ and $p=x^{4000}+\frac{1}{x^{4000}}$ and q is the digit at unit place in the number $2^{2^n}+1, n\in N$ and $n>1$, then the value of $p+q=8$.
Statement II: ω,ω^2 are the roots of $x+\frac{1}{x}=-1, x^3+\frac{1}{x^3}=2$

Which of the above statement(s) is/are correct ?
(a) Only I (b) Only II
(c) Both I and II (d) Neither I nor II

133. If $\omega(\neq 1)$ is cube root of unity satisfying $\frac{1}{a+\omega}+\frac{1}{b+\omega}+\frac{1}{c+\omega}=2\omega^2$ and $\frac{1}{a+\omega^2}+\frac{1}{b+\omega^2}+\frac{1}{c+\omega^2}=2\omega$, then the value of $\frac{1}{a+1}+\frac{1}{b+1}+\frac{1}{c+1}$ is equal to
(a) 2 (b) -2
(c) $-1+\omega^2$ (d) None of these

134. If α and β are the roots of the equation $x^2-2x+4=0$, then the value of $\alpha^6+\beta^6$ is
(a) 64 (b) 128
(c) 256 (d) None of these

135. If z is any complex num ch that $z+\frac{1}{z}=1$, then the value of z^9
(a) 1
(c) 2

136. The val $+\sqrt{-3})^{62}+(-1-\sqrt{-3})^{62}$ is
(a) 2^{62} (b) 2^{64}
(c) -2 (d) 0

s a cube root of unity, then
$\omega+3\omega^2)^2(3+3\omega+5\omega^2)^2$ is equal to
(b) 0
c) -4 (d) None of these

138. The value of $\left(\frac{1+i\sqrt{3}}{1-i\sqrt{3}}\right)^6+\left(\frac{1-i\sqrt{3}}{1+i\sqrt{3}}\right)^6$ is
(a) 2 (b) -2
(c) 1 (d) 0

139. If $z_1=\sqrt{3}+i\sqrt{3}$ and $z_2=\sqrt{3}+i$, then the complex number $\left(\frac{z_1}{z_2}\right)^{50}$ lies in the quadrant number
(a) I (b) II
(c) III (d) IV

140. If $x=a+b, y=a\omega+b\omega^2, z=a\omega^2+b\omega$, then xyz is equal to
(a) $(a+b)^3$ (b) a^3+b^3
(c) a^3-b^3 (d) $(a+b)^3+3ab(a+b)$

141. If $1,\omega$ and ω^2 are the cube roots of unity, then the value of $(1+\omega)^3-(1+\omega^2)^3$ is
(a) 2ω (b) 2
(c) -2 (d) 0

142. If $1,\omega$ and ω^2 are the three cube roots of unity and α,β and γ are the roots of $p, p<0$, then

for any x, y and z, the expression $\frac{x\alpha+y\beta+z\gamma}{x\beta+y\gamma+z\alpha}$ equals
(a) 1 (b) ω
(c) ω^2 (d) None of the above

143. If α and β are the complex cube roots of unity, then $\alpha^3+\beta^3+\alpha^{-2}\beta^{-2}$ is equal to
(a) 0 (b) 3
(c) -3 (d) None of these

144. If ω is a complex cube root of unity, then $(1-\omega+\omega^2)(1-\omega^2+\omega^4)(1-\omega^4+\omega^8)$ $(1-\omega^8+\omega^{16})$ is equal to
(a) 12 (b) 14
(c) 16 (d) None of these

145. If $x=\omega-\omega^2-2$, then the value of $x^4+3x^3+2x^2-11x-6$ is
(a) 1 (b) -1
(c) 2 (d) None of these

146. If $1, \omega$ and ω^2 are the three cube roots of unity, then $(1+\omega)(1+\omega^2)(1+\omega^4)(1+\omega^8)$ ... to $2n$ factors is equal to
(a) 1 (b) -1
(c) 0 (d) None of these

147. The common roots of the equations z^3 $2z^2+2z+1=0$ and $z^{1985}+z^{100}+1$
(a) $-1, \omega$ (b) $-1, \omega^2$
(c) ω, ω^2 (d) None of the

148. The values of $(16)^{1/4}$ a
(a) $\pm 2, \pm 2i$ (b) $\pm$
(c) $\pm 1, \pm i$ (d) e above

149. If p, q and r are e integers and ω is an imaginary cube ity and $f(x)=x^{3p}+x^{3q+1}+x^{3r+2}$, $\omega)$ is equal to
(a) ω (b) $-\omega^2$
(c) 0 (d) None of the above

150. If $\omega(\neq 1)$ is a cube root of unity and $(1+\omega)^7=A+B\omega$. Then, (A, B) equals
(a) $(1,1)$ (b) $(1,0)$
(c) $(-1,1)$ (d) $(0,1)$

151. If α and β are the roots of the equation $x^2-x+1=0$, then $\alpha^{2009}+\beta^{2009}$ is equal to
(a) -2 (b) -1
(c) 1 (d) 2

152. If $z^2+z+1=0$, where z is complex number, then the value of $\left(z+\frac{1}{z}\right)^2+\left(z^2+\frac{1}{z^2}\right)^2+\left(z^3+\frac{1}{z^3}\right)^2+\cdots+\left(z^6+\frac{1}{z^6}\right)^2$ is
(a) 54 (b) 6
(c) 12 (d) 18

153. If the cube roots of unity are $1, \omega, \omega^2$, then the roots of the equation $(x-1)^3+8=0$, are
(a) $-1, 1+2\omega, 1+2\omega^2$
(b) $-1, 1-2\omega, 1-2\omega^2$
(c) $-1,-1,-1$
(d) $-1,-1+2\omega,-1-2\omega^2$

154. If ω is an imagin oot of unity, then $(1+\omega-\omega^2)^7$ eq
(a) 128ω -128ω
(c) $128\omega^2$ (d) $-128\omega^2$

155. If he cube roots of unity, then roots of equ $-1)^3+25=0$ are
, $\omega, -5\omega^2$
$,1-5\omega, 1-5\omega^2$
$,1-5\omega, 1+5\omega^2$
d) None of the above

156. The value of $(1+\sqrt{3}i)^4+(1-\sqrt{3}i)^4$ is
(a) -16 (b) 16
(c) 14 (d) -14

157. If $\left(\frac{3}{2}+i\frac{\sqrt{3}}{2}\right)^{50}=3^{25}(x+iy)$, where x and y are real, then the ordered pair (x, y) is
(a) $(-3,0)$ (b) $(0,3)$
(c) $(0,-3)$ (d) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

158. If a and b are real numbers such that $(2+\alpha)^4=a+b\alpha$, where $\alpha=\frac{-1+i\sqrt{3}}{2}$, then $a+b$ is equal to
(a) 24 (b) 33
(c) 9 (d) 57

159. The value of $\left(\frac{-1+i\sqrt{3}}{1-i}\right)^{30}$ is
(a) -2^{15} (b) $2^{15}i$
(c) $-2^{15}i$ (d) 6^5

160. If $i=\sqrt{-1}$, then $4+5\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)^{334}+3\left(-\frac{1}{2}+i\frac{\sqrt{3}}{2}\right)^{365}$ is equal to
(a) $1-i\sqrt{3}$ (b) $-1+i\sqrt{3}$
(c) $i\sqrt{3}$ (d) $-i\sqrt{3}$

161. $\sqrt{-1-\sqrt{1-\sqrt{-1-\cdots\infty}}}$ is equal to
(a) 1 (b) -1
(c) ω^2 (d) $-\omega$

"Locus of Complex Number"

162. The set of values of k for which the equation $z\bar{z}+(-3+4i)\bar{z}-(3+4i)z+k=0$ represents a circle is
(a) $(-\infty,25)$ (b) $(25,\infty)$
(c) $(5,\infty)$ (d) $(-\infty,5)$

163. The complex numbers $z=x+iy$ which satisfy the equation $\left|\frac{z-5i}{z+5i}\right|=1$, lie on
(a) the X-axis
(b) the straight line $y=5$
(c) a circle passing through the origin
(d) None of the above

164. If the equation
$|z-z_1|^2+|z-z_2|^2=k$ represents the
of a circle, where $z_1=2+3i, z_2=4+$ e
extremities of a diameter, then the f k is
(a) $\frac{1}{4}$ (b) 4
(c) 2 (d) N ese

165. The locus of z g $\frac{1}{z-i}\Big|=1$ is
(a) a circle an ellipse
(c) a straight li (d) a parabola

166. If $z=x+iy$ and
$|z-2+i|=|z-3-i|$, then locus of z is
(a) $2x+4y-5=0$ (b) $2x-4y-5=0$
(c) $x+2y=0$ (d) $x-2y+5=0$

167. The equation

$$|z-i|+|z+i|=k, k>0$$

can represent an ellipse, if k^2 is
(a) <1 (b) <2
(c) >4 (d) None of these

168. The equation $|z+i|-|z-i|=k$ represents a hyperbola if
(a) $-2<k<2$ (b) $k>2$
(c) $0<k<2$ (d) None of these

169. $\left|\frac{z-1}{z+1}\right|=1$, represents
(a) a circle (b) an ellipse
(c) a straight line (d) N these

170. $|z-4|<|z-2|$, e region given by
(a) Re $(z)>0$ $(z)<0$
(c) Re $(z)>$ None of these

171. If z $\frac{}{2}$ then z is represented by a point l
(a) a c (b) an ellipse
tr ht line (d) None of these

If z is a complex number, then $z^2+\bar{z}^2=2$ epresents
a) a circle (b) a straight line
(c) a hyperbola (d) an ellipse

173. The equation $|z+1-i|=|z+i-1|$ represents
(a) a straight line (b) a circle
(c) a parabola (d) a hyperbola

174. If z_1, z_2 and z_3 are three complex numbers in AP, then they lie on
(a) a circle (b) a straight line
(c) a parabola (d) an ellipse

175. If Re $\left(\frac{z+2i}{z+4}\right)=0$, then z lies on a circle with centre
(a) $(-2,-1)$ (b) $(-2,1)$
(c) $(2,-1)$ (d) $(2,1)$

176. If Im $\left(\frac{z+2i}{z+2}\right)=0$, then z lies on the curve
(a) $x^2+y^2+2x+2y=0$
(b) $x^2+y^2-2x=0$
(c) $x+y+2=0$
(d) None of the above

177. If $z = x + iy$ and a is a real number such that $|z - ai| = |z + ai|$, then locus of z is
(a) X-axis (b) Y-axis
(c) $x = y$ (d) $x^2 + y^2 = 1$

178. If P represents $z = x + iy$ in the argand plane and $|z-1|^2 + |z+1|^2 = 4$, then the locus of P is
(a) $x^2 + y^2 = 2$ (b) $x^2 + y^2 = 1$
(c) $x^2 + y^2 = 4$ (d) $x + y = 2$

179. The locus of z satisfying Im $(z^2) = 4$ is
(a) a circle
(b) a rectangular hyperbola
(c) a pair of straight lines
(d) None of the above

180. The curve represented by Re $(z^2) = 4$ is
(a) a parabola (b) an ellipse
(c) a circle (d) a rectangular hyperbola

181. If $w = \frac{z}{z - \frac{i}{3}}$ and $|w| = 1$, then z lies on
(a) a parabola (b) a circle
(c) a straight line (d) an ellipse

182. The complex number $z = x + iy$, which satisfies the equation $\left|\frac{z-3i}{z+3i}\right| = 1$, lie on
(a) the X-axis
(b) the straight line $y = 3$
(c) a circle passing through the origin
(d) None of the above

183. If $\left|\frac{z-2}{z-3}\right| = 2$ represents en centre and radius are
(a) $\left(\frac{10}{3}, 0\right), \frac{2}{3}$ ($0), \frac{}{3}$
(c) $\left(\frac{10}{3}, 1\right), \frac{2}{3}$ None of these

"Minimu d Maximum Value"

184. For a complex number z, the minimum value of $|z| + |z-2|$ is
(a) 1 (b) 2
(c) 3 (d) None of these

185. If $|z_1 - 1| < 1, |z_2 - 2| < 2$ and $|z_3 - 3| < 3$, then $|z_1 + z_2 + z_3|$
(a) is less than 6 (b) is more than 3
(c) is less than 12 (d) lies between 6 and 12

186. The maximum and minimum values of $|z + 1|$, when $|z + 3| \leq 3$ are
(a) (5,0) (b) (6,0)
(c) (7,1) (d) (5,1)

187. The minimum value of $|1 + z| + |1 - z|$, where z is a complex number, is
(a) 2 (b) $\frac{3}{2}$
(c) 1 (d) 0

188. If $\left|z + \frac{4}{z}\right| = a$, then find the greatest and least values of $|z|$.

189. If z is any compl ch that $|z + 4| \leq 3$, then the lue and greatest value of $|z + 1|$ are
(a) 1,6 0,6
(c) 2,8 (d) None of these

190. F omplex number z, the minimum value $|z - 1|$ is
(b) 0
(d) $\frac{3}{2}$

91. The maximum value of $|z|$ when z satisfies the condition $\left|z + \frac{2}{z}\right| = 2$ is
(a) $\sqrt{3} - 1$ (b) $\sqrt{3} + 1$
(c) $\sqrt{3}$ (d) $\sqrt{2} + \sqrt{3}$

192. If $|z_1| = 15$ and $|z_2 - 3 - 4i| = 5$, then which of the following is correct ?
(1) $|z_1 - z_2|_{\min} = 5$ (2) $|z_1 - z_2|_{\min} = 10$
(3) $|z_1 - z_2|_{\max} = 20$ (4) $|z_1 - z_2|_{\max} = 25$

193. If $|z - (1/z)| = 1$, then which of the following statement is correct ?
(1) $|z|_{\max} = \frac{1+\sqrt{5}}{2}$ (2) $|z|_{\min} = \frac{\sqrt{5}-1}{2}$
(3) $|z|_{\max} = \frac{\sqrt{5}-2}{2}$ (4) $|z|_{\min} = \frac{\sqrt{5}-1}{\sqrt{2}}$

194. If z is a complex number such that $|z| \geq 2$, then the minimum value of $\left|z + \frac{1}{2}\right|$
(a) is equal to $\frac{5}{2}$
(b) lies in the interval (1,2)
(c) is strictly greater than $\frac{5}{2}$
(d) is strictly greater than $\frac{3}{2}$ but less than $\frac{5}{2}$

195. If $\left|z-\frac{4}{z}\right|=2$, then the maximum value of $|z|$ is equal to

(a) $\sqrt{3}+1$ (b) $\sqrt{5}+1$
(c) 2 (d) $2+\sqrt{2}$

196. The complex number z satisfies the condition $\left|z-\frac{25}{z}\right|=24$. The maximum distance from the origin of coordinates to the point z is

(a) 25 (b) 30
(c) 32 (d) None of these

Two Important Points :

(i) The area of triangle whose vertices are z, iz and $z+iz$ is $\frac{1}{2}|z|^2$.

(ii) The area of triangle whose vertices are $z, \omega z$ and $z+\omega z$ is $\frac{\sqrt{3}}{4}|z|^2$.

Example :

197. If the area of the triangle on the complex plane formed by the points z, iz and $z+iz$ is 50 units, then $|z|$ is

(a) 5 (b) 10
(c) 15 (d) None of these

198. If the area of the triangle on t lex plane formed by complex n and $z+$ ωz is $4\sqrt{3}$ sq units, then $|z|$ i

(a) 4 (b)
(c) 6 (